Guidelines

Evidence collection, packaging and submission guidelines by discipline.

I. Digital Evidence

Collection and Packaging

- Evidence must be packaged in sealable container of appropriate size that also prevents contamination or deterioration.
- Package evidence in a manner to prevent bending, scratching, or other damage. Avoid the use of plastic materials for the purpose.
- Evidence collected must be properly documented, labelled, marked, photographed and inventoried prior to packaging.
- The evidence item may also contain latent, trace, or biological evidence; take appropriate measures to preserve it.
- Package digital evidence in anti-static containers such as paper bags, envelopes, or cardboard boxes to protect it from static electricity.
- Video evidence should be collected in its original format as it is recorded on the recording device (DVR etc.).
- Ensure that the Pin, pattern or password of the digital device (mobile phone etc.) is provided by the evidence submitting person or agency.
- Collect all power supplies, cables, and adapters associated with seized electronic devices.
- Use shock-resistant packaging to protect all components of the device(s) from physical damage.
- Label all containers used to package digital evidence clearly and properly.
- Ensure packaging area is free from ultraviolet (UV) light, such as from certain fluorescent tubes, as UV exposure can accelerate evidence degradation.
- The packaging environment should have a moderate temperature and humidity. An extreme environment can lead to spoliation of potential evidence.
- Each seal should display both the initials of the seal creator and the date it was applied.
- Use tamper-evident seals that clearly show any attempt at removal or interference.
- Evidence packaging must include, at minimum, the submitting agency case number, item number, date, and the initials of the person who packaged it.
- It is advisable that forensic evidence tape is used to seal the evidence.
- Leave mobile devices in the power state in which they are found; if possible, enable flight or airplane mode.
- Isolate mobile devices from networks using techniques such as Faraday bags, radio frequency shielding materials, anti-static packaging, or aluminum foil.

Storage

- Make sure that the digital evidence is properly inventoried.
- Store digital evidence in a secure, climate-controlled environment, avoiding extreme temperature or humidity.

- Protect evidence from magnetic fields, moisture, dust, vibration, or any other conditions that could cause damage.

Transportation

- Keep evidence away from magnetic fields (e.g., radio transmitters, speaker magnets, magnetic-mount emergency lights) and other hazards like seat heaters or materials that may generate static electricity.
- Do not leave evidence in vehicles for extended periods, as heat, cold, or humidity can damage or destroy it.
- Ensure that electronic devices/computers are packaged and secured during transportation to prevent damage from vibration and shock.
- Document the transportation of the evidence and maintain the chain of custody for every evidence transported.

II. Finger Prints

General Handling / Collection

- Personnel must wear appropriate Personal Protective Equipment (PPE) such as disposable gloves and face masks when handling latent fingerprint evidence.
- Latent fingerprints can be developed directly at the crime scene by crime scene investigation personnel using suitable techniques depending on the surface involved. Latent prints developed through conventional powder methods should first be photographed, then lifted using transparent lifting tape, and subsequently submitted to the laboratory. Case details, date, location, and orientation of the developed print should be recorded on the lift card.
- If fingerprints are visible to the naked eye (patent prints) at crime scene, or if lifting the print presents difficulty, photographic documentation should be performed. However, when an item is intended to be sent to the laboratory for processing, attempts to recover latent prints in the field should be avoided.
- Non-porous surfaces (e.g., glass, metal etc.) may be processed at the crime scene using fingerprint powders, preferably black powder for better ridge detail and comparison. Powder processing requires a dry surface; wet items should either be treated with Small Particle Reagent (SPR) or allowed to air-dry naturally. Whenever possible, such items should be processed at the crime scene, with the developed prints photographed, lifted, and submitted to NFA for further enhancement and comparison.
- Excessive handling and unnecessary transportation may damage print(s). In certain situations, Cyanoacrylate Ester (commonly known as super glue fuming) may be used for developing latent prints, particularly on materials such as plastic bags, firearms, Styrofoam, and some types of leather.
- Porous or absorbent surfaces (e.g., paper, untreated wood, cardboard) are generally processed using chemical methods such as Indanedione, Ninhydrin, Physical Developer for wet surfaces, and Amido Black for bloody fingerprints; chemically developed prints must be photographed as they may fade or disappear over time.
- Bloody fingerprints should be photographed with scale before and after applying any blood enhancement technique (e.g., Amido Black).

- For visible prints on small movable objects (e.g., a cup), collect the entire object as evidence; if the item is fixed and cannot be transported, it should be processed at the crime scene for latent fingerprint development.
- Do not press or touch the impression with fingers or any object to check whether the substance is dry or sticky, as this may damage the print.
- Photographs are essential, as attempts to enhance or lift the print may cause damage to the original impression.

Packaging

- Non-porous items should be packaged in such a way that they remain in fixed position and should not move freely inside the package during handling and transportation. Use of cardboard boxes and plastic ties is recommended as packaging material.
- Dry paper items may be collected and sealed in plastic (zip-lock) bags, whereas wet paper items should be air-dried completely before packaging.
- Sharp objects, such as broken glass or knives, must be safely packaged and clearly labeled. Paper bags should not be used, as sharp items can puncture them and cause injury.

Case Submission

- Clearly indicate all requested forensic examinations on the Evidence Submission Form; for resubmissions, include the previous NFA case number.
- Do not process items intended for laboratory submission, and avoid placing tape over areas that may contain latent prints.
- High-quality known prints are essential; smudged, blurred, over-inked, off-center, or overlapping prints may reduce identification accuracy.
- If suspects are known, obtain and submit their fingerprint and palm print cards along with the evidence.
- Collect elimination prints from victims, family members, caretakers, or others who may have had legitimate contact with the evidence.
- Submit the original questioned document for Fingerprint Examination.
- Docket / Cover letter addressed to the Director General, NFA, Islamabad clearly indicating required analysis and details of questioned and reference thumb impressions shall be required.

Case fee shall be charged for all cases as per the NFA Fee Schedule. Payment shall be made via Bank Draft or Pay Order in favor of DG, NFA, Islamabad (photocopy of bank draft is not acceptable), or through any other mode of payment as approved by the competent authority. For criminal cases, a copy of FIR shall also be submitted.

III. Firearms and Tool Marks

Collection

- Each evidence exhibit should be collected and packaged separately.
- Firearms must be unloaded and the safety engaged.
- Ensure no live rounds are present in the chamber, magazine, or within the parcel.
- Every cartridge case and bullet must be collected individually.

- For GSR analysis, clothing layers must not touch each other; place a white paper sheet between layers before packaging in a hard box.
- For serial number restoration of firearms, clearly mark areas with obliterated serial numbers if more than one location exists.
- For trajectory analysis, vehicles must not be washed or cleaned; suspected bullet holes should be covered with white paper.
- If a firearm is recovered from water or another liquid, submit it along with the same liquid sample.

Storage

- Ensure that all seals on evidence remain intact and match those documented in the docket.
- Protect fragile evidence such as serial numbers, bullet holes, and GSR samples from damage or contamination.

Transportation

- Use appropriate containers for each type of evidence (hard boxes for firearms and GSR, separate parcels for bullets and cartridge cases).
- If the evidence is related to a prior submission, clearly reference its link to the earlier case.
- Handle all evidence carefully during transport.

IV. Polygraph Examination

Following documents are required for polygraph examination

- Original CNIC of the nominated individual along with photocopy/Passport/Form B in case of minors.
- In case the original CNIC is missing, a Missing Identity Form issued by the relevant authority must be provided.
- Copy of FIR
- Request letter for polygraph examination

Following conditions must be met; otherwise, the polygraph examination will not be administered

- The suspect has taken breakfast and have appropriate sleep.
- Individual is not seriously ill or injured.
- No intoxicants have been taken in the last 24 hours.
- Individual has not been informed beforehand about the polygraph examination.
- No physical assault or harassment or mistreatment has been done in the last week.
- All documents/letters are less than one week old.
- No female suspect shall be accompanied by children.
- Individual has washed hands with soap.
- Investigating officer has studied the case properly.

V. Forensic Pathology

Forensic pathology evidence includes biological specimens such as tissues, organs, bones, teeth, and body fluids collected during autopsy, exhumation, or medico-legal examination. Proper collection, preservation, and transportation of these specimens are essential to prevent decomposition, contamination, or loss of evidentiary value. The following guidelines outline the recommended procedures to ensure that pathology samples are handled securely, preserved adequately, and submitted to the NFA in a condition suitable for accurate examination and analysis.

Collection

Tissue Samples:

- Required specimens should be collected immediately following autopsy or exhumation to minimize degradation.
- Each specimen must be placed in a container containing freshly prepared 10% formalin, ensuring complete immersion of the tissue.
- The volume of formalin should be 3-4 times the volume of the tissue to ensure proper fixation.
- Record essential details at the time of collection, including specimen type, anatomical site, date and time of sampling, and name and designation of the collector.
- For cytological specimens such as body fluids, secretions, or blood, add a few drops of 10% formalin in the specimen.

Exhumation Cases:

- Fixative has to be added even in exhumation cases.
- Only the necessary specimens should be collected to reduce excessive handling and minimize the risk of contamination.

Preservation and Packaging

Tissue Samples:

- Use clean plastic jars with screw-cap lids for tissue specimens.
- Ensure the tissue is fully immersed in 10% formalin, maintaining the recommended fixative-to-tissue ratio.
- Close lids tightly and seal the container using evidence tape; if unavailable, red wax seals may be used as an alternative.
- Place sealed containers in plastic bags or secondary protective packaging to prevent leakage during handling and transport.

Fluids and Small Specimens:

- Store liquids or small specimens in sterile, leak-proof containers.
- Clearly label each container with case number, specimen description, date and time of collection, and initials of the collector.

- Place containers within secondary containment (sealed plastic bags or rigid boxes) to prevent accidental spillage.

Fragile or Sharps Items:

- Bones, teeth, or other sharp fragments should be padded or secured to prevent damage to the specimen or injury to handlers.
- Avoid direct contact between sharp objects and other specimens within the same package.

Labeling of samples:

Mention following information on the label on sample jars:

- Name of the deceased
- PMR/Case number
- Sample details
- Date and time of sampling
- Collectors name, designation and signature

Transportation

- Place sealed specimen containers upright in appropriately sized cardboard boxes, ensuring they remain stable and do not move during transport.
- Clearly mark the package with "This Side Up" to maintain proper orientation.
- Seal all box openings with evidence tape, bearing the signature and official stamp of the responsible officer.

Documentation

All relevant documents must accompany the samples, including:

- Postmortem report
- FIR / Application Rapat
- Medico-Legal Certificate (MLC)
- Road Certificate
- Concise case details (Mukhtasar Halat e Muqadma)
- Court or Magistrate order in exhumation cases (legible copy)
- Relative/family request
- Put all the relevant documents along with sample of evidence tape and/or sample of signatures and stamp in an envelope.
- Seal the envelope with evidence tape signature and stamp as described above. If evidence tape is not available, stamped red wax seal may be used alternatively, as described above.
- Send the histopathology samples and documents to NFA.

VI. Questioned Documents

Collection & Packaging

- Package questioned document evidence in a suitably sized paper envelope without folding the documents.
- Record all necessary information on the envelope before placing the evidence inside; do not write on the envelope after packaging.
- If the evidence is to be tested for indented writing or latent fingerprints, ensure it is packaged carefully so it does not rub against other documents.
- Attach all necessary case-related documents.

Preservation

- Use envelopes that protect the documents from wear, tear, and contamination.
- Protect documents from harsh environmental conditions such as moisture or fire.
- For charred or water-damaged documents, use a hardboard box or suitable container with cotton cushion to prevent further damage.

Transportation

- Submit or dispatch the evidence in a properly sealed condition.
- Docket or cover letter should be addressed to the Director General, National Forensics Agency, Islamabad, clearly stating the required examination.
- Clearly mark and mention questioned, routine/admitted, and dictated exemplars in the docket/cover letter.
- Submit the case fee via Bank Draft or Pay Order or through other approved mode of payment in favor of the Director General, National Forensics Agency, Islamabad.

Note: For cases with a large number of questioned exhibits, only probative items, as determined with the investigating officer, will be examined.

VII. Explosive Evidence (Pre-blast/Post blast)

General Considerations

- 2-5 gram of sample from the suspected explosive material/bomb device should be received.
- For detonating cords like prima cord or safety fuse, submit only 3–6 inches of the cord.
- Intact detonators, hand grenades, suicide jackets, or mines are strictly prohibited in the lab.
- Only small portions of explosives (2–5 g; for detonators 0.1–0.5 g) should be submitted in properly labeled, sealed parcels.
- Evidence must be delivered to NFA by an authorized person from the submitting agency; courier submission of trace evidence is not allowed.
- Debris from blast scenes (soil, metal fragments, plastic, ball bearings, jagged fabric, broken timing devices, etc.) must be packaged in separate, sealed, labeled containers with the collection location clearly noted.
- Submit an intact "control" soil sample from an uncontaminated area near the blast scene in a separate sealed and labeled container.

Packaging & Sealing Requirements

- The packaging container must be unused and airtight.
- It must be clean, with no hydrocarbon or chemical residues.
- It must be inert i.e. should not degrade when heated or exposed to solvents.
- It should not promote a static electrical charge (important for explosives).
- Use clean, intact seals; containers must be fully sealed to prevent vapor or contaminant escape.
- Plastic bags must be heat sealed with no flaws when used for fire debris or explosive sample.
- Tamper-evident tape must cover container seams so items cannot be opened without tearing the tape.
- Seal and tape must be dated, and include the case number, exhibit number, and investigator's initials/signature for record.

VIII. Trace Chemistry

Trace Chemistry Department deals with a wide range of materials (hair, fibers, paint, adhesive tapes, acids, arson debris, GSR, footwear/tire impressions). Each crime scene must be carefully examined, and all probative evidence should be collected carefully to prevent contamination or loss, preserving the integrity of the samples. Evidence must be delivered to NFA by an authorized person from the submitting agency; courier submission of trace evidence is not allowed.

General Principles

- Each container must be labeled with: Case number, exhibit number, item description, date, and collector's initials.
- Handle evidence carefully to avoid contamination, damage, or degradation.
- Different types of evidence require specific collection and packaging methods.

Evidence Collection Methods

Trace evidences like hair, fiber, paint chips, adhesive tapes etc. can be collected by the following methods:

I. Handpicking

- Use forceps or an appropriate tool to gently lift evidence from the surface.
- Place the evidence in a suitable container and seal.

II. Tape lifting

- Remove a few inches of clean tape from the roll to remove contamination.
- Cut a tape section appropriate for the item.
- Fold ends to create handles.
- Apply the adhesive side to the item to collect trace evidence.
- Transfer to a storage backing (e.g., acetate sheet) and seal in a container.

III. Shaking

- Place clean examination paper under the object.
- Shake the object gently to release trace material.
- Any evidentiary material observed on the examination paper is removed by handpicking.
- The debris is transferred to a sealed container.

IV. Scraping

- Place a section of examination paper under the item to be examined.
- Scrap the evidence item with a clean scraping tool over the section of examination paper.
- Visually examine the section of examination paper for the presence of evidentiary material. Handpick any evidentiary material observed on the examination paper.
- The debris on the section of examination paper is transferred to a container and sealed.

Paint Evidence

- Paint samples collected should represent all the layers of the paint present. The sample should be chipped off down to the unpainted surface.
- Whenever possible, submit the entire object on which the paint is observed, including smears and transfers. Do not attempt to remove paint from clothing, tools or objects where smears and transfers are deposited.
- If it is not feasible to submit the entire object, use a clean knife blade or scalpel to remove the area of interest including all the layers possible.
- Small samples can be collected using forceps or tweezers.
- Place sample in a paper fold or vial. Do not use an envelope. Small samples may be lost among the folds, openings and seals of the envelope.
- Place different samples in separate containers to avoid contamination.
- Be sure to seal the container and record the proper identifying information on the container and exterior packing.

Reference Comparison Sample for Paint Analysis:

A reference paint sample is a known, undamaged paint collected from the same object or area as the damaged paint to serve as a comparison during analysis.

- Reference paint samples should measure at least ½ square inch and include all paint layers down to the substrate.
- Collect reference paint samples from areas adjacent to the damage, as paint composition may differ across panels or sections of a vehicle or object, even if the color looks identical. Ensure a standard is taken from each separate panel or area showing fresh damage.
- Store each reference paint sample in a separate paper fold, then properly seal and label it.
- The label should include case and investigator information, as well as the exact source of the paint (e.g., vehicle make and model). Collect reference paint from all vehicles or objects involved, even if some known paint is included with the questioned paint.

Hair Evidence

The Trace Chemistry Department examines hair evidence to assess its potential for DNA profiling. Evaluation includes determining whether the hair is human or animal, identifying the body origin (e.g., scalp, pubic), and noting the growth phase.

Hair evidence can be collected in a number of ways including the following methods:

- Picking (For visible hair)
- Tape lifts
- Scraping
- Shaking

If the entire object, such as an article of clothing, containing possible hair evidence is to be submitted to the lab, place the object onto clean craft paper and paper fold. Seal the fold and place in a paper bag or envelope. Seal the container and include the proper identifying information.

Hair Sample Standards

- Roots: Always include the hair roots, as they provide critical information.
- Head/Scalp Hair: Collect approximately 50 hairs from different areas of the scalp (center, front, back, nape, and both sides). Include both pulled and combed hairs, covering variations in color and length. Facial hair (e.g., beard, sideburns) should be collected and packed separately.
- Pubic Hair: Collect 20–30 hairs when relevant to the case.
- Animal Hair: Hair should be combed and pulled to include roots, which are needed for species identification. Collect sufficient hair to represent different types and colors, from multiple areas (head, back, belly, tail, etc.). Pack each sample separately and label it with the body area of collection.

Fiber Evidence

Fiber evidence may be collected in the same manner as hair evidence. These methods include but not limited to picking, tape lifts, and scraping.

- Collect fiber standards from all surfaces or items that the victim and suspect may have contacted.
- Whenever possible, submit the entire item as a fiber standard. If this is not feasible, cut a representative swatch (e.g., car seat) or pull random fibers (e.g., carpet).
- When collecting fiber standards from a vehicle, collect samples from all the areas where fibers may have transferred (i.e., front and rear floorboards, carpeting, mats, upholstery and trunk liners). Even if areas look similar, fibers may differ by location, and lab analysis may be required to distinguish them.

Important: Do not place loose fibers directly into envelopes. Always use a paper fold for collection.

Adhesive Tape

- Tape fragments recovered from the crime scene or victim should be placed in separate, sealed, and properly labeled parcels, preferably affixed on plastic sheets.

- Reference tape fragments/rolls collected from the suspect should be submitted in separate, sealed and labeled parcels for comparison purposes.

Acid Examination

- The suspected acid container recovered from the scene or from the possession of the suspect should be properly sealed and packaged in an airtight, leak-proof container.
- Collect approximately 15–20 ml of the suspected liquid in a sealed and labeled airtight container.
- Affected clothing of the victim should be packed in a sealed and labeled airtight container.
- Collect debris from the scene suspected of containing acid residues in clean, sealed, and labeled airtight containers, clearly indicating the exact location of collection.
- Biological samples (skin, flesh, human remains, blood etc.) are not accepted in Trace Chemistry Department.

Fire/Arson Cases

- Any suspected ignitable liquid container recovered from the scene or from a suspect should be made airtight or placed inside a clean, airtight metal container (e.g., a paint can).
- Collect approximately 15–20 ml of suspected ignitable liquid in a sealed and labeled airtight container.
- Semi-burnt clothing of the victim should be packaged in a sealed, labeled, airtight metal container.
- Debris from the scene suspected of containing ignitable liquid residues should be collected in clean, sealed, and airtight metal containers, properly labeled.
- Collect a control sample (e.g., soil, intact carpet) from an unaffected area for comparison.
- All packaging materials must be unused, clean, and free from contaminants (e.g., hydrocarbons or chemical residues).
- Use containers that are airtight, inert, and resistant to heat or solvents, such as metal cans or heat-resistant nylon bags.

Gunshot Residue (GSR)

- Collect GSR using pure carbon adhesive stubs from both hands of the shooter, covering the backs and palms, and place them in a sealed and labeled parcel.
- At least two adhesive stubs should be collected: one from each hand (back and palm areas).
- Cotton swabs or hand washes are not acceptable for GSR analysis.
- Sampling should be conducted within 4–6 hours of the shooting, as residue may no longer be detectable after this period.

Footwear/Tire Impression

- Footwear impressions (molds/casts) recovered from the scene should be packaged in sealed and labeled cardboard boxes.

- Reference molds of the suspect's footwear should be submitted, preferably along with the actual shoes, in a separate sealed and labeled cardboard box.
- Tire track mold/cast from the place of occurrence should be packaged in a sealed and labeled cardboard box.
- Reference tire molds should be submitted, preferably along with the vehicle tires, for comparison.
- Bare foot impressions are not accepted in Trace Chemistry Department.

Packaging & Sealing Requirements

- The packaging container must be unused and airtight.
- It must be clean, with no hydrocarbon or chemical residues.
- It must be inert i.e. should not degrade when heated or exposed to solvents.
- It should not promote a static electrical charge (important for explosives).
- Use clean, intact seals; containers must be fully sealed to prevent vapor or contaminant escape.
- Plastic bags must be heat sealed with no flaws when used for fire debris or explosive sample.
- Tamper-evident tape must cover container seams so items cannot be opened without tearing the tape.
- Seal and tape must be dated, and include the case number, exhibit number, and investigator's initials/signature for record.

Evidence Packaging Instructions

Packaging Type	Case Type / Use
Paper folds	Fibers, paint chips, glass particles (minute), hair
Metal cans or nylon bags	Arson-related evidence, such as burnt clothing
Cardboard boxes	Cases of physical match: knives, glass pieces (large), plastic, vehicle paint from clothing
Pure carbon adhesive stub	Primer gunshot residue collection
Glass vials	Suspected powder/ignitable liquids/acids/bases/explosive

IX. Narcotics

Narcotics-related evidence include controlled drugs, psychotropic substances, precursor chemicals, and associated materials such as plant matter, powders, liquids, and drug paraphernalia. Such evidence is highly susceptible to contamination, loss, and legal challenges; therefore, strict adherence to proper collection, secure packaging, accurate labeling, and documented chain of custody is essential to preserve its integrity and ensure admissibility in forensic analysis and court proceedings.

- Evidence must be packaged in sealed, clean containers such as plastic bag, cloth wrap, paper envelope or box.
- Liquid samples must be packaged in leak proof, sealed bottle.
- All samples must be packaged separately.

- Wet plant material should be placed in paper envelope or paper fold to prevent deterioration.
- Seal markings on the evidence must match those stated in the FIR.
- FIR copy, Analysis Request Letter (e.g., from DPO/CPO letter or equivalent) and Road Certificate etc. should be submitted along with the evidence.
- The court order for analysis is the minimum requirement for the cases submitted by courts.
- FIR number and evidence description should be consistent across all the documents and evidence packaging.
- Evidence should be submitted by the Investigation Officer or by the person named on the Road Certificate as the submitting person.

X. DNA & Serology

Biological evidence includes materials of biological origin such as blood, semen, saliva, hair, tissue, bones, teeth, and other body fluids, as well as any items containing such materials. The purpose of proper handling of biological evidence is to:

- Ensure safety of personnel handling the material
- Preserve the integrity and quality of DNA
- Prevent contamination and degradation

General Safety Guidelines

- All biological materials must be treated as potentially infectious, as the presence of bloodborne pathogens (e.g., hepatitis, HIV) cannot be determined in every specimen.
- Use appropriate Personal Protective Equipment (PPE) including gloves, masks, protective clothing, and eye protection to prevent exposure to biological hazards.
- Change gloves between handling different items to prevent cross-contamination.
- PPE must be clean, intact, and single-use; contaminated or damaged PPE must be replaced immediately.
- Avoid touching face or exposed skin while handling evidence.
- Remove PPE before leaving the work area and dispose of it in biohazard-designated containers.
- Handle both wet and dried biological materials with equal caution.
- In case of spills, disinfect the area using 10% bleach solution, leave for at least 10 minutes, and dispose of waste safely.
- Clearly label all packages as "BIOLOGICAL EVIDENCE".
- Any accidental exposure must be reported immediately.

Collection of Biological Evidence

- Collect each item carefully and separately to prevent cross-contamination.
- The biological evidence should be air-dried before packaging, as moisture promotes microbial growth and DNA degradation.
- Wet items such as bloodstained clothing, swabs, or biological stains must be dried in a clean, controlled environment.
- Avoid using heat sources for drying, as excessive heat may damage DNA.

Collection of Biological Evidence in Sexual Assault Cases

Sexual Assault Evidence Collection Kits (SAECK) shall be used for standardized and effective collection of biological evidence in sexual assault cases. These kits contain sterile swabs, distilled water, combs, gloves, packaging materials, and tamper-evident seals to ensure proper collection, preservation, and submission of evidence.

Pre-Collection Considerations:

Evidence collection should be conducted after obtaining a detailed history and medico-legal examination (MLC), documenting:

- Nature of assault
- Time elapsed since assault
- Post-assault activities (e.g., bathing, urination, oral hygiene)
- Age and gender of the assault victim
- Relevant personal and medical details of the victim
- Collection strategy should be guided by case history and physical findings.
- In cases where history is incomplete (e.g., trauma or incapacity), a comprehensive range of samples should be collected based on examination findings.

Collection of Evidence:

- Collect undergarments and clothing worn during the incident and package them in the provided envelopes.
- Collect both internal and external swabs (vaginal, anal, oral, skin surfaces), even if the victim has bathed, as biological material may still persist.
- Items such as towels, tissues, bedding, or other substrates must always be collected, as biological evidence on such materials may persist longer than on the body.
- Use only sterile cotton swabs (provided in SAECK or commercially available).
- Do not use improvised materials such as cotton balls or homemade swabs.

Swab Collection Technique:

- Use the minimum number of swabs necessary to concentrate foreign DNA.
- Where multiple swabs are required, prefer concurrent collection, or clearly document the order of collection.
- Maintain consistency in technique (e.g., if one swab is moistened, all should be similarly prepared).

Post-Collection Handling:

- All swabs must be air-dried thoroughly before packaging.
- Swabs must never be placed in liquid or preservative.
- Wet or bloodstained garments must also be air-dried prior to packaging.

Time Frames for Evidence Collection:

Biological evidence persistence varies by site. Recommended maximum collection intervals are:

Assault Type	Maximum Post-Coital Time for Evidence Collection
Oral	24 hours
Anal	72 hours
Bite marks	96 hours
Vaginal	120 hours

Packaging of Biological Evidence

Packaging Material:

- Use porous materials such as paper bags, manila envelopes, and cardboard boxes for packaging biological evidence.
- Do not use non-porous materials (e.g., plastic bags, glass bottles, metal containers, polythene), as they retain moisture and promote bacterial growth and mold, leading to degradation of biological material.

Labelling Requirements:

Each package must be clearly and legibly labelled with:

- "BIOLOGICAL EVIDENCE" (prominently displayed)
- Unique identifier corresponding to case records (FIR, MLC, PMR, etc.)
- Name/identifier of the patient
- Description of the evidence item
- Date of collection
- Name and signature/initials of the collector

XI. Toxicology

Sample must be submitted in preservative and amount as described below:

For Medico-legal cases (MLCs)

- Blood: 10 mL blood preserved with mixture (ratio 1:3) of sodium fluoride and potassium oxalate. 20 mg of this mixture is sufficient for preservation of 10 mL blood.
- Urine: 20 mL urine without any preservative.
- Gastric Lavage: Minimum 20 mL, first undiluted portion of gastric lavage without preservative.

For Postmortem cases

- Blood: 50 mL blood preserved with mixture (ratio 1:3) of sodium fluoride and potassium oxalate. 100 mg of this mixture is sufficient for preservation of 50 mL blood.
- Urine: All available urine without any preservative.
- Gastric contents: All available gastric contents without any preservatives.
- Liver: Not more than 20 grams liver preserved in saturated saline.

- Spleen: If carbon monoxide poisoning is suspected and blood is not available, then 20 g spleen preserved in saturated saline.
- Abdominal paste: If above mentioned samples are not available in exhumation case, then 20 g abdominal paste without any preservative.
- Hair: Accepted only in chronic drug/poison exposure (Hair cluster having thickness of a pencil, pulled or collected as near to scalp as possible).

Collection, Preservation and Transport of Evidence

- Collect toxicology samples as soon as possible after the incident, in death cases prior to embalming (where applicable).
- Ensure that the specimens are packaged in tightly sealed, leak-proof containers.
- All samples must be collected in separate containers. For most specimens, disposable hard plastic or glass tubes are recommended.
- Blood tubes should be sealed and kept cold (but do not freeze). Do not expose specimens to hot temperatures.

Labeling

To maintain a valid chain of custody, each evidence item must be clearly labeled with:

- Name of victim or suspect.
- Case number.
- Type of specimen (i.e. urine/blood).
- Site of collection (i.e., heart/femoral).
- Amount of specimen.
- Time and date of collection.
- Name(s) of the medical examiner or person collecting the sample.

Place tamper-resistant tape over the specimen lid and container, including the collector's initials and date, to ensure integrity. As an alternative, all samples collected for a given case may be placed in a tamper-evident container (labelled with the case number & name).